

1	Course Name:
	Sensor Technology for IoT
2	Course Code:
	IBB22803
3	Year and Semester Offered:
	Year 2 Semester 2
4	Credit Hours Value:
	3 Credit Hours [3 SLT Credit]
5	Prerequisite(s), if applicable:
	IBB22503 Electronics
6	Course Synopsis:
	This course covers topics to the introduction of sensors, examples of nature sensors, type of sensors/sensors in different application areas and sensor networks. Topics include sensor classification, characteristics, principles of sensing, electronic interface, electronic/integration, introduction to microsystem and its components and sensor materials and technologies.
7	Course Learning Outcome:
	Upon completion of the course, students should be able to: 1. Assess the right sensor for a given application. 2. Appraise system using selected sensors. 3. Assemble system using the selected sensors.
8	Content Outline:
	Topic
	1. Introduction to Sensors Example of Sensor in nature (Vision, Hearing, Touch and Smell) How to learn from Nature 2. Sensors: Principles, Classification and Terminology 3. Characteristics of Sensors, Sensor Network Discussion on electronic field, potential etc. Sensor materials and Technologies Electronic Interface and Integrated Sensors 4. MEMS Component, Introduction to Microsystems Electronic and wireless integration sensors 5. Application: - Sensors in different application areas and for practical application (PBL) - Sensor Materials and Technologies